CHIRANJIBEE PANIGRAHI

AWS CLOUD & DEVOPS ENGINEER | INFRASTRUCTURE AUTOMATION

Bhubaneswar, Odisha, India • **chiranjibeepanigrahi@gmail.com** • **LinkedIn** • **GitHub: Chiranjibee18**

TECHNICAL SKILLS

Cloud (AWS):	VPC (Peering, Route Tables, Subnets, IGW), EC2, IAM (Least Privilege), S3, RDS, Lambda, CloudWatch
Containers & K8s:	Docker, Multi-stage Dockerfiles, Kubernetes (Deployments, Services, Ingress, HPA, ConfigMaps, Secrets, Probes)
CI/CD & IaC:	Jenkins (Declarative Pipelines), GitHub Actions, Git/GitHub, Terraform (Modules, State Locking), Ansible
Monitoring & OS:	Prometheus, Grafana, CloudWatch Metrics/Alarms, Linux Administration (Ubuntu, Amazon Linux), Bash Scripting
Networking & Security:	TCP/IP, DNS, SSH ProxyJump, Security Groups, CIDR Subnetting, ICMP, Non-transitive Routing, SSL/TLS

ENGINEERING PROJECTS

AWS VPC Peering & Secure Cross-Network Architecture

View on GitHub ↗

Stack: AWS VPC, Route Tables, Security Groups, EC2, SSH ProxyJump, ap-south-1

- **Multi-VPC Isolation:** Architected dual-VPC network (`10.0.0.0/16` & `10.1.0.0/16`) in Mumbai region, isolating backend application compute from internet access while establishing point-to-point VPC Peering (`pcx`).
- **Bidirectional Routing & Latency:** Configured symmetrical route tables and explicit subnet associations, achieving private, cross-network ICMP communication with **0% packet loss** and sub-millisecond round-trip time (**0.738 ms**).
- **Keyless Bastion Administration:** Implemented client-side OpenSSH **ProxyJump** via a public jump host, eliminating credential theft risks by preventing sensitive private keys (`.pem`) from residing on cloud instances.
- **Least-Privilege Security:** Enforced granular `/32` IP whitelisting across Security Groups; performed controlled failure testing by removing ICMP ingress rules to validate stateful security group enforcement.

Kubernetes Microservices Platform — Scaling & Observability

View on GitHub ↗

Stack: Kubernetes, Docker, Helm, Prometheus, Grafana, HPA

- **Production Workload Management:** Deployed containerized microservices using Kubernetes Deployments and ClusterIP Services, decoupling application rollout from internal service discovery.
- **Autoscaling & Self-Healing:** Configured **Horizontal Pod Autoscaler (HPA)** based on CPU/memory utilization; implemented **Liveness and Readiness probes** to guarantee zero traffic routing to unready or failing pods.
- **Secure Configuration & Routing:** Configured Ingress for HTTP routing and external exposure; decoupled sensitive credentials and environment variables using Kubernetes Secrets and ConfigMaps.
- **Full-Stack Observability:** Deployed Prometheus metric scraping with Grafana visualization, establishing dashboards for pod restarts, CPU saturation, and cluster resource bottlenecks.

Automated CI/CD Pipeline & Containerized Deployment

View on GitHub ↗

Stack: Jenkins, Docker, GitHub Webhooks, Linux, Bash Scripting

- **Automated Delivery:** Built an end-to-end declarative Jenkins CI/CD pipeline triggered on GitHub push/PR events to automate build, testing, container packaging, and deployment stages.
- **Image Optimization:** Authored multi-stage Dockerfiles that minimized container image size by **~45%** and removed build-time dependencies from production runtimes.
- **Operational Reliability:** Automated Docker container lifecycle, volume mounts, and port bindings on Linux hosts, reducing release downtime and manual deployment overhead.

Terraform AWS Cloud Infrastructure as Code (IaC)

View on GitHub ↗

Stack: Terraform, AWS (VPC, EC2, IAM, S3), Git

- **Declarative Provisioning:** Automated provisioning of repeatable multi-tier AWS environments (VPCs, public/private subnets, Security Groups, EC2) using reusable Terraform modules and parameterized variables.
- **State & Lifecycle Management:** Managed state lifecycle using `terraform plan`, `apply`, and `destroy` workflows; established consistent environment separation for dev and staging.

EDUCATION

Institute of Technical Education and Research (ITER), SOA University — B.Tech in Computer Science & Engineering — **CGPA: 8.65 / 10** | 2024 – 2028

Senior Secondary (Class XII): 85.4% | Secondary Education (Class X): 90.8% — Bhubaneswar, Odisha

CERTIFICATIONS & HACKATHONS

Certifications: KodeKloud (Linux Administration & Security, Docker & Kubernetes Orchestration, AWS Fundamentals) | Cisco (Python Essentials, Cybersecurity Basics)

Achievements: Selected Participant — **Smart India Hackathon (SIH)** | Presented Cloud Deployment Showcase at **Nirmaan Tech Expo**